

The right page is usually **found**, but rarely **first**.

How the system answers a question today:

- 1 Trained Hebrew model turns the question into a meaning vector
- 2 Elasticsearch returns the **50 closest paragraphs**
- 3 Collapse to pages. A page ranks by its **single best paragraph**

Step 3 is the weak spot. A page with *one* paragraph that happens to look like the question can outrank the page that is actually *about* the question.

FULL CORPUS · 24,487 PARAGRAPHS · BASELINE

Correct page in the **top 10**

87%

Correct page at **#1**

39%

The right page is already among the candidates for almost 9 questions in 10, but only the top few pages reach the answer step. **That gap is the target.**

Score the **page**, not just its luckiest paragraph.

```
page score = best paragraph
+ 0.25 × second-best paragraph
+ 0.25 × title match
```

All three from the **same trained model**. A small margin gate (0.05) keeps generic "hub" pages from jumping up from far below.

Second-best paragraph

The right page usually has several matching paragraphs. A look-alike has one.

Title match

Kol-Zchut titles name the topic. Paragraphs are fragments that often don't.

עד מתי מקבלים קצבת ילדים? → קצבת ילדים

no new model

no re-indexing

one flag, off = original

milliseconds per query

WHY NOT THE OBVIOUS UPGRADES?

The shipped retriever was **trained on these very questions**. Anything that overrules it loses. Both ideas were built, integrated and tested the same way.

IDEA	HELD-OUT #1, %
Cross-encoder reranker (BGE) re-reads and re-sorts the 50	45 → 40
Page scoring listens to the same model more carefully	39 → 55

Held-out sets differ: 100 questions for the reranker, 250 fresh ones for page scoring (on those same 100 it was 45 → 50). The rejected idea helped ranks 4–5 and hurt the first result.

First result: 38.8% → 54.8% on questions the weights never saw.

CORRECT PAGE IS...	BEFORE	AFTER	CHANGE
at #1	38.8%	54.8%	+16.0
in top 3	65.6%	71.6%	+6.0
in top 5	77.2%	78.4%	+1.2
in top 10	86.8%	87.6%	+0.8
MRR@10 (how high up, on average)	0.547	0.650	+0.10

Metrics: hit@k because #1 is what the user and the answer step actually see. MRR because it rewards moving the right page *upward*. Same 50 candidates before and after, only the order changes, so top-5 and top-10 are essentially unchanged (within noise): reordering fixes rank, not recall.

HOW IT WAS KEPT HONEST

Weights chosen on **250** questions, reported **once** on 250 held-out ones. Ship rule for every idea: held-out #1 or MRR up, top-5 not down. One of two failed it.

IN THE BACKEND

An optional step in the engine's search, turned on in the local demo. Off restores the original exactly. No API key needed. 9 unit tests, plus a check that the live API returns the offline ranking.

HONEST LIMITS

The QA file also trained the retriever, so absolute numbers are optimistic. The split guards against tuning, not familiarity. The title weight relies on good titles.